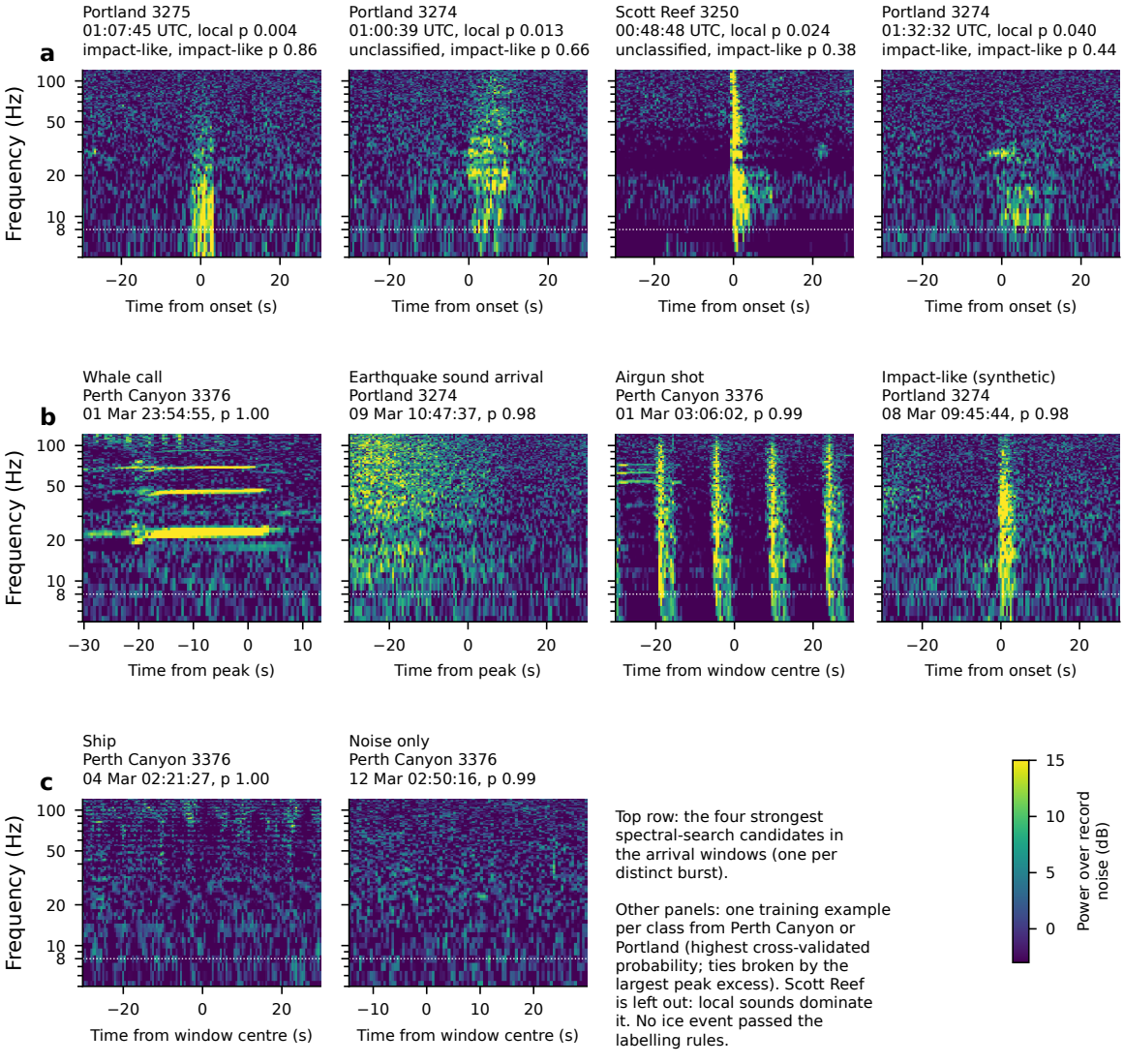
Spectral-search candidates and one example of each source class



Spectrograms: IMOS logger records at 250 Hz, 1 s frames, 0.5 s step, power divided by the record's own noise spectrum (quietest 80 % of frames). Content below 8 Hz (dotted line) is removed by the logger's hardware filter. Search: impact (source) time 00:19:38-00:29:44 UTC; 7th arc 20-41 S, +/-50 NM; sound speed 1.40-1.56 km/s plus up to 5 % detour and a 300 s land-path branch. Propagation is used for timing only. Template family (fixed before running; post-hoc in origin): loose impact class, slope flat to -15 dB/octave, corner 0.5-20 Hz, band cut-offs 8-20 Hz and 15-120 Hz, at least one octave wide, multipath ripple, rise 0.25-3 s, decay 0.5-30 s; the Metz et al. aircraft-impact shape is one member. Level is free. Local p is against 280 background-day records (Perth Canyon 3376, Scott Reef, Portland 3274) or same-day off-window records (Perth Canyon 3315, Portland 3275). No candidate reaches station-global p of 0.05 (best 0.19, Portland 3275). The Portland 3275 burst at 01:07:35-45 UTC also matches the predicted arrival of Royer event 6 (class uncertain; association probability 0.75). It is kept. Known events: removed only at combined probability 0.95 or more with positive class evidence; none removed. The synthetic impact example is regenerated with its training parameters (new noise draw). The earthquake example rests on catalogue timing; about half of such labels may be chance matches.